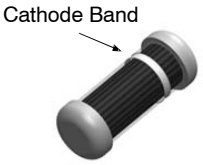
High Conductance, Low Leakage Diode

FDH300 / FDH300A / FDLL300A / FDH333 / FDLL333



AXIAL LEAD
(DO−35)
CASE 017AG

Cathode is denoted with a black band



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(The placement of the expansion gap has no relationship to the location of the cathode terminal)

ABSOLUTE MAXIMUM RATINGS

(T_A = 25°C unless otherwise noted) (Note 1) (Note 2)

Symbol	Parameter	Value	Units
W_{IV}	Working Inverse Voltage	125	V
I_O	Average Rectified Current	200	mA
I_F	DC Forward Current	500	mA
i_f	Recurrent Peak Forward Current	600	mA
I_{FSM}	Non−Repetitive Peak Forward Surge Current Pulse Width = 1.0 s Pulse Width = 1.0 μs	 1.0 4.0	A
T_{STG}	Storage Temperature Range	−65 to +200	°C
T_J	Operating Junction Temperature	175	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. These ratings are based on a maximum junction temperature of 175°C.
2. These are steady−state limits. **onsemi** should be consulted on applications involving pulsed or low−duty−cycle operations.

THERMAL CHARACTERISTICS

Symbol	Parameter	Value	Units
P_D	Total Device Dissipation	500	mW
	Derate Above 25°C	3.33	mW/°C
$R_{\theta JA}$	Thermal Resistance, Junction to Ambient	300	°C

MARKING DIAGRAM

LL−34 COLOR BAND MARKING

DEVICE	1ST BAND
FDLL300A	WHITE
FDLL300A	WHITE

−1st band denotes cathode terminal and has wider width

ORDERING INFORMATION

See detailed ordering, marking and shipping information in the package dimensions section on page 2 of this data sheet.

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ELECTRICAL CHARACTERISTICS (Values are at T_A = 25°C unless otherwise noted.)

Symbol	Parameter		Condition	Min	Max	Unit
V_R	Breakdown Voltage		I_R = 100 μA	150	–	V
V_F	Forward Voltage	FDH300 / FDH300A / FDLL300A	I_F = 1.0 mA	–	680	mV
		FDH300	I_F = 5.0 mA	–	750	mV
		FDH300A / FDLL300A	I_F = 5.0 mA	–	760	mV
		FDH300 / FDH300A / FDLL300A	I_F = 10 mA	–	800	mV
		FDH300	I_F = 50 mA	–	880	mV
		FDH300A / FDLL300A	I_F = 50 mA	–	890	mV
		FDH300 / FDH300A / FDLL300A	I_F = 100 mA	–	920	mV
		FDH300 / FDH300A / FDLL300A	I_F = 200 mA	–	1.0	V
		FDH333 / FDLL333	I_F = 50 mA	800	890	mV
			I_F = 100 mA	830	940	mV
			I_F = 150 mA	860	970	mV
			I_F = 200 mA	0.87	1.05	V
			I_F = 250 mA	0.88	1.08	V
			I_F = 300 mA	0.90	1.15	V
I_R	Reverse Current	FDH300 / FDH300A / FDLL300A	V_R = 125 V	–	1.0	nA
			V_R = 125 V, T_A = 150°C	–	3.0	μA
		FDH333 / FDLL333	V_R = 125 V	–	3.0	nA
			V_R = 125 V, T_A = 100°C	–	500	nA
C_O	Diode Capacitance		V_R = 0, f = 1.0 MHz	–	6.0	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

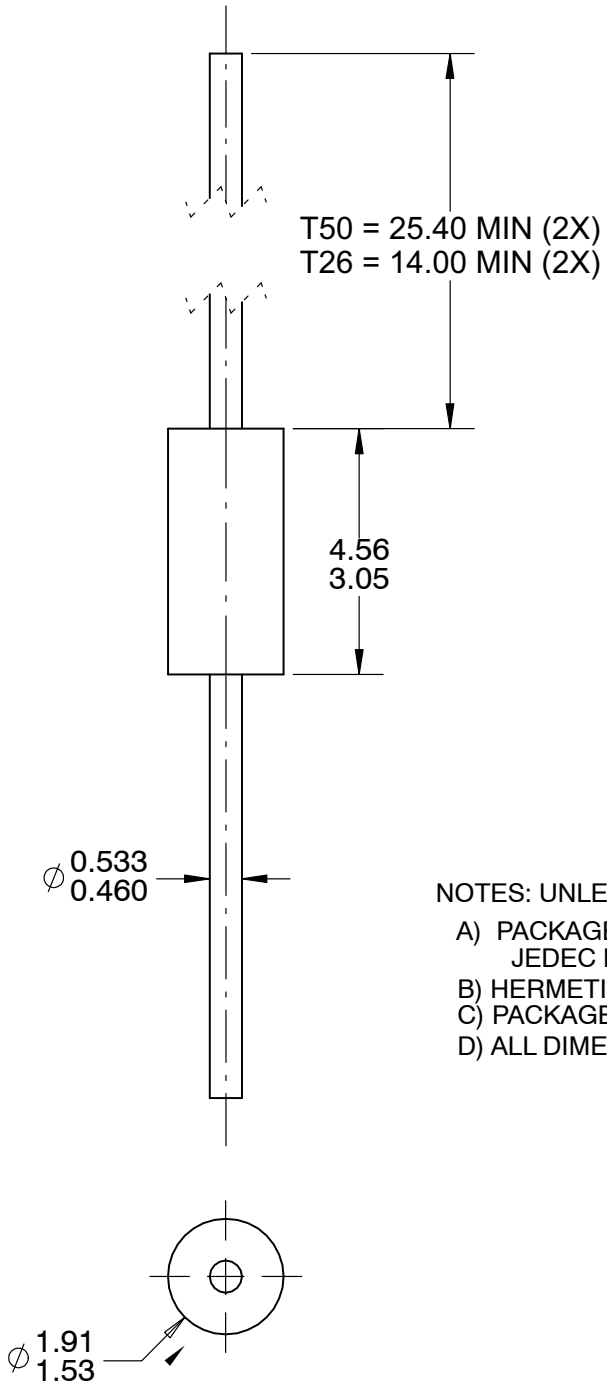
DEVICE ORDERING INFORMATION

Part Number	Top Mark	Package	Shipping†
FDH300TR	H300	DO−204AH (DO−35)	Tape and Reel
FDH300A	H300A	DO−204AH (DO−35)	Bulk
FDH300ATR	H300A	DO−204AH (DO−35)	Tape and Reel
FDH333	H333	DO−204AH (DO−35)	Bulk
FDH333TR	H333	DO−204AH (DO−35)	Tape and Reel
FDLL300A	WHITE	SOD−80 2L	Tape and Reel
FDLL333	WHITE	SOD−80 2L	Tape and Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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DATE 31 AUG 2016



NOTES: UNLESS OTHERWISE SPECIFIED

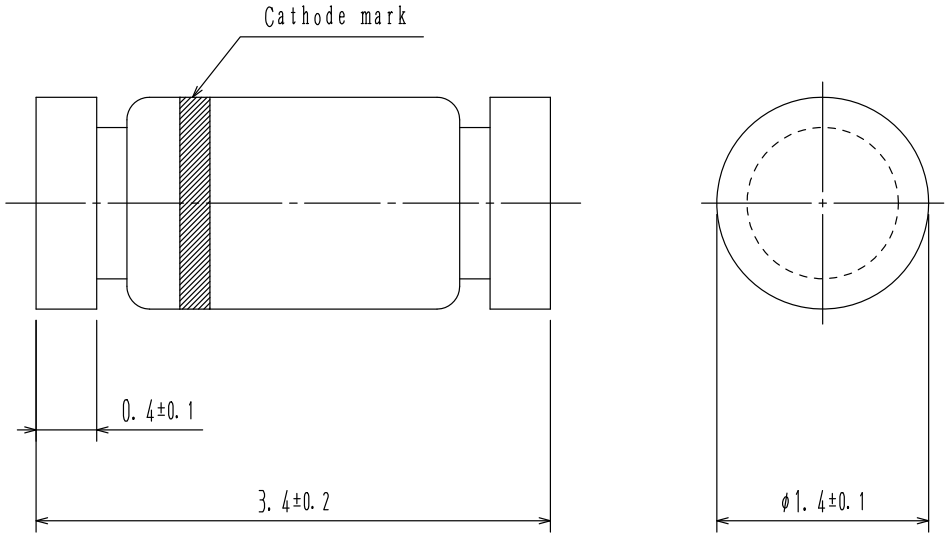
A) PACKAGE STANDARD REFERENCE:
JEDEC DO−204, VARIATION AH.
B) HERMETICALLY SEALED GLASS PACKAGE.
C) PACKAGE WEIGHT IS 0.137 GRAM.
D) ALL DIMENSIONS ARE IN MILLIMETERS.

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CASE 100AD
ISSUE O

DATE 30 APR 2012



NOTES: UNLESS OTHERWISE SPECIFIED

A) PACKAGE STANDARD REFERENCE: JEDEC DO-213, VARIATION AC.

B) ALL DIMENSIONS ARE IN MILLIMETERS.

△C CORNER RADIUS IS OPTIONAL.

D) DRAWING FILE NAME: SOD80A REV01

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